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EMM: Plug-and-play memory-bank sampling for contrastive recommendation

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ABSTRACT

In contrastive recommendation, representation quality depends not only on the encoder but also on sample construction: positives drive alignment and negatives enforce uniformity, shaping gradient strength, optimization stability, and data efficiency under sparse feedback. However, two key challenges have not been well explored in existing solutions: (i) how to obtain negatives that are simultaneously informative, fresh, and diverse, since random sampling yields weak gradients while static hardest only mining quickly becomes stale; (ii) how to safeguard against false negatives, where semantically close items are mistakenly penalized, distorting geometry and destabilizing training. To cope with these challenges, we propose Episodic Memory Mining (EMM), which reframes negative sampling from a brittle heuristic into a principled component for contrastive recommendation, remaining plug-and-play by intervening only at the sampling layer and keeping backbones intact. EMM keeps negatives fresh, informative, and diverse via an EMA memory with periodic rebuild, Top-K retrieval from a refreshed candidate pool, and a small random-negative mix with rolling refresh to stabilize gradients. Across three benchmarks and five backbones, EMM yields about 15.1% average improvement in overall recommendation quality at comparable training cost and shows broader hyperparameter robustness. This supports negative sample construction as a simple and general lever for stronger contrastive recommendation under sparse feedback.

1. Introduction

In recent years, graph-based collaborative filtering has converged toward simple yet strong backbones paired with self-supervised contrastive objectives. LightGCN exemplifies this trend by reducing graph convolution to linear neighborhood aggregation and has become a standard backbone for top-K recommendation (He et al., 2020b). Building on such backbones, SGL introduced contrastive learning on recommender graphs by maximizing agreement between two views of the same node to alleviate sparsity (Wu et al., 2021), and subsequent methods (e.g., SimGCL, XSimGCL, and LightGCL) further showed that performance is often determined less by architectural complexity than by how training samples are constructed, including the choice of views and, crucially, the negatives used in the contrastive denominator (Cai et al., 2023; Yu et al., 2023, 2022). This line of work suggests that improving contrastive recommendation increasingly hinges on principled sample construction, setting the stage for a closer look at the remaining limitations of negative sampling in sparse-feedback settings.

Nevertheless, important limitations persist and they motivate a more systematic design. First, forming positives by coupling two noisy graph augmentations can remove informative edges and shift semantics; empirical results show that mild representation-level noise or globally regularized views can be more stable than heavy perturbations (Cai et al., 2023; Yu et al., 2023, 2022). Second, deeper propagation causes over-smoothing, which homogenizes node embeddings across layers and weakens the semantic anchors that contrastive objectives rely upon; theory clarifies conditions under which limited smoothing is beneficial and conditions under which excessive smoothing harms discrimination (Keriven, 2022; Oono & Suzuki, 2020). Third, negative sampling remains fragile: purely random negatives provide weak gradients, whereas mining only the hardest neighbors risks false negatives and training instability. Debiased ob-

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jectives, hard-negative mixing, and counterfactual hard negatives have been proposed to strengthen gradients while improving safety and stability, yet these techniques are often adopted in isolation and without a unifying recipe tailored to recommendation scenarios (Chuang et al., 2020; Kalantidis et al., 2020; Yang et al., 2023).

To address these limitations, we introduce Episodic Memory Mining (EMM), a model-agnostic plugin standardizes negative construction as a drop-in approximation of the contrastive denominator and can be attached to SGL, SimGCL, and LightGCL pipelines without changing the backbone encoder or forward propagation. While EMM draws on memory-bank and hard-negative mining ideas, its novelty lies in how these components are composed into a single, recommender-specific negative formation primitive that is simultaneously fresh, informative, and safe under sparse feedback. Concretely, EMM maintains an EMA-updated memory bank with periodic episodic rebuild to counter representation drift, and performs Top-K hard-negative mining on a refreshed candidate pool, rather than a static hardest-only set, to retain informative gradients at low cost. To stabilize optimization and sustain diversity, EMM samples negatives from a controlled hard–random mixture, and applies rolling candidate refresh to keep the pool current. Crucially, EMM also incorporates explicit false-negative safeguards: (i) ID exclusion to prevent paired positives/self IDs from entering the negative set, and (ii) a similarity window that filters overly close candidates as well as overly distant ones, implemented via masked logits without changing the loss interface (Chuang et al., 2020; He et al., 2020a; Johnson et al., 2017; Kalantidis et al., 2020; Yang et al., 2023). Together, these design choices turn negative sampling from an ad hoc heuristic into a plug-and-play training primitive that improves stability and data efficiency while preserving the original geometry of LightGCL-style contrastive objectives.

This work makes four contributions:

(1) We propose EMM, which recasts the full-library log-sum-exp denominator as a refreshable, memory-driven candidate approximation that can be plugged into existing contrastive recommenders without modifying backbones or positive construction.

(2) We introduce a two-timescale episodic refresh mechanism (EMA updates plus periodic rebuild) together with rolling candidate refresh, mitigating stale hard negatives and reducing sensitivity to sampling hyperparameters.

(3) We design stability- and safety-aware negative formation via a controlled hard–random mixture and explicit false-negative safeguards (ID exclusion and similarity-window filtering), yielding stronger and more reliable gradients than random sampling or hardest-only mining.

(4) We empirically demonstrate consistent improvements on Baby, Sports, and Clothing across five representative baselines at comparable training cost, and provide ablations isolating the effect of refresh, mixture sampling, and safeguards, supporting the view that negative construction is a first-class and orthogonal design axis for contrastive recommendation.

2. Related work

Self-supervised recommenders and view construction. Recent studies move beyond generic perturbations and emphasize principled view formation tailored to recommendation scenarios. For spatiotemporal mobility and next-POI tasks, disentangled or hypergraph formulations learn multiple behavioral facets and contrast them to improve discrimination under sparse signals (Lai et al., 2024). In sequential recommendation, SelfGNN builds short-term interaction graphs and applies self-supervision over interval-aware subgraphs, while ICSRec contrasts cross-subsequences to expose latent intents without relying on noisy random augmentations (Liu et al., 2024; Sheng et al., 2024). Temporal contrast is made explicit by TGCL4SR, which aligns local and global temporal signals within a unified objective (Zhang et al., 2024). For cold-start cross-domain recommendation, DisCo proposes a graph-based disentangled contrastive framework that captures fine-grained user intents and filters irrelevant source-domain signals to mitigate negative transfer (Li et al., 2025).Beyond deterministic transformations, meta-optimized joint generative and contrastive training integrates a generator with a contrastive learner to produce semantically consistent positives that better preserve sequence semantics (Zhou et al., 2024). More recently, LLM-enhanced paradigms such as DCRLRec reinforce representations in both collaborative and semantic-graph domains and align them via contrastive reinforcement, indicating the growing interest in combining semantic reasoning with contrastive objectives for recommendation (Bai et al., 2025).

Alignment, uniformity, and depth robustness. A parallel thread examines representation geometry and the role of depth in contrastive optimization. Decision boundary–aware graph contrast constrains augmentation within semantically safe regions and reduces representation drift during training (Tang et al., 2024). Kernel dependence maximization replaces mutual-information surrogates with a non-parametric dependence criterion to stabilize alignment without over-tight clustering, offering an alternative to vanilla InfoNCE for social graphs (Ni et al., 2024). On the sequence side, coordinating multiple contrast signals within one objective improves generalization, and disentangling long- and short-term interests under self-supervision preserves low-layer semantics instead of encouraging indiscriminate smoothing (He et al., 2025; Wang et al., 2023). While most prior work enforces positive-side geometry or replaces the objective, our approach targets the negative side by reshaping the denominator via a memory-banked Top-K mix sampler, which improves the alignment-uniformity trade-off and exhibits reduced depth sensitivity.

Negatives: hardness, diversity, and sampling policy. Performance depends on negatives that are informative but safe. PopDCL introduces a debiased contrastive loss that is based on popularity that corrects the sampling bias for positives and negatives, and the popularity-based alignment further reweights contrast to mitigate head-item dominance (Cai et al., 2024; Liu et al., 2023). Beyond reweighting, trainable negative generators learn adversarial negatives that adapt to the current representation space (Zhuang et al., 2024). At larger scales, expansive yet controlled subspace preservation widens the search region for hard negatives while limiting drift, and contrastive clustering for multibehavior recommendation balances hardness with diversity when multiple feedback types coexist (Hao et al., 2024; Lan et al., 2024). We follow this direction with a per-view EMA memory bank, Top-K retrieval, and a small random mix. The bank and periodic rebuild ensure freshness and stability, while the mix and exclusion rules maintain diversity and safety.Importantly, we distinguish EMM from recent generative hard-negative approaches. Generative methods typically synthesize challenging negatives via an explicit generator or LLM-driven semantic reasoning, whereas EMM is retrieval-based and performs episodic mining from a refreshed candidate pool supported by an EMA-updated memory bank and efficient Top-K search. This design keeps EMM plug-and-play at the sampling layer without introducing an additional generative model, and enables an explicit freshness–cost trade-off through episodic rebuild and rolling refresh. Moreover, EMM can incorporate generated hard negatives as additional candidates in the pool, and then apply the same safety filtering and refresh strategy to select stable and informative negatives.

Positioning of this work. Previous efforts often optimize positives through intent-aware, temporal, or generative views, or negatives through debiasing and adaptive mining, with method-specific code paths that are difficult to port. We instead provide a plug-and-play sample constructor that is backbone-agnostic. Our novelty lies in standardizing negative formation as a plug-and-play sampling primitive that jointly controls hardness, freshness, stability, and false-negative safety under sparse implicit feedback. EMM operates purely at the sampling layer and can be attached to SGL/SimGCL/LightGCL-style pipelines without modifying backbones, forward propagation, or loss interfaces.

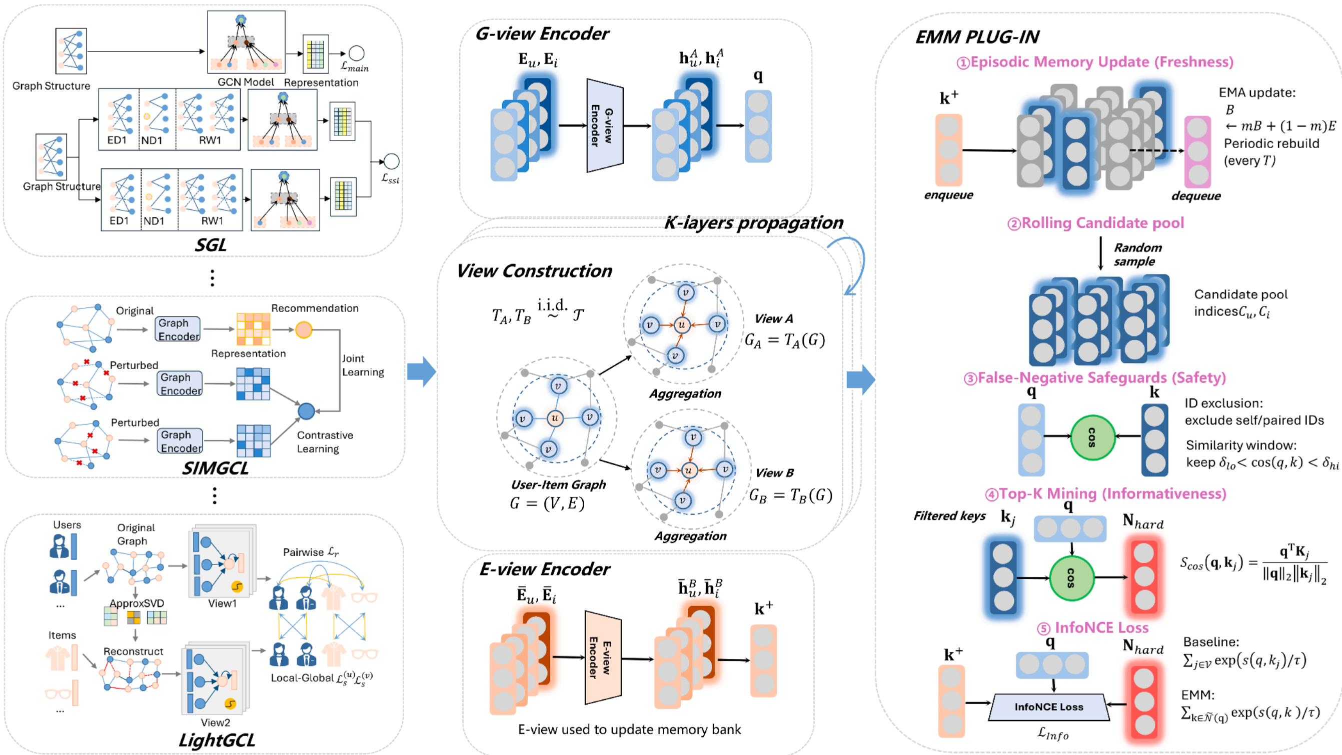


Fig. 1. Overall framework of EMM. EMM keeps the backbone and positive construction unchanged and replaces the full-library contrastive denominator by a refreshed negative set It maintains an episodically refreshed memory bank, refreshes a candidate pool, mines Top-K negatives with a small random mix, and applies ID exclusion plus a similarity-window filter to mitigate false negatives before computing InfoNCE.

3. Methodology

This section introduces the details of our plug-and-play *Episodic-Memory Mining (EMM)* module. Fig. 1 shows the overall architecture of EMM. It comprises four components: (1) Dual-view encoding that produces an anchor q via the online encoder f_θ and a positive key k^+ via the momentum encoder $\bar{f}_\theta$, enabling robust positive alignment with EMA-updated targets and enqueueing; (2) Episodic memory bank, a queue of historical keys that mixes the current mini-batch with buffered keys to form a candidate pool; (3) Dynamic hard-negative mining that first suppresses false negatives and then selects informative negatives via stratified Top-k/segment sampling to build N_{hard}; and (4) Joint training objective that optimizes contrastive InfoNCE over $(q, k^+, N_{\text{hard}})$ together with a standard BPR loss, updating f_θ while $\bar{f}_\theta$ follows EMA. This design is model-agnostic and can be integrated into other models that are based on contrastive-learning architectures with minimal changes.

3.1. Problem setup and notation

We consider a user–item bipartite graph $G = (\mathcal{U}, \mathcal{I}, \mathcal{E})$ with $|\mathcal{U}| = M$ users, $|\mathcal{I}| = N$ items, and an implicit–feedback edge set $\mathcal{E} \subseteq \mathcal{U} \times \mathcal{I}$. The backbone encoder maintains trainable tables $\mathbf{E}_u \in \mathbb{R}^{M\times d}$ and $\mathbf{E}_i \in \mathbb{R}^{N\times d}$ and, after its own propagation, outputs top-layer representations $\mathbf{Z}_u \in \mathbb{R}^{M\times d}$ and $\mathbf{Z}_i \in \mathbb{R}^{N\times d}$. We distinguish these from clean (non-augmented) embeddings $\widetilde{\mathbf{Z}}_u$ and $\widetilde{\mathbf{Z}}_i$ obtained from a forward pass without stochastic perturbations; the latter are used solely to maintain memory banks. For contrastive learning, each entity $x \in \mathcal{U} \cup \mathcal{I}$ has two baseline views yielding an anchor $\mathbf{z}_x^{(a)}$ and a positive $\mathbf{z}_x^{(p)}$. We keep the positive construction unchanged across backbones. Let $s(\cdot,\cdot)$ be the similarity function and $\tau > 0$ the temperature.

Training uses the standard multi-task objective:

$$\mathcal{L} = \mathcal{L}_{\text{main}} + \lambda\,\mathcal{L}_{\text{ssl}} + \beta\,\|\Theta\|_2^2,$$

where Θ collects all learnable parameters and $\lambda, \beta \geq 0$. Throughout, we do not modify the backbone's forward propagation or $\mathcal{L}_{\text{main}}$; our plug-in affects only the negative set inside the contrastive denominator, while positives remain as in the baseline. For batching, we write $\mathcal{B}_u \subseteq \mathcal{U}$ and $\mathcal{B}_i \subseteq \mathcal{I}$, and denote by $\mathcal{N}(\mathbf{z}^{(a)})$ the negative set associated with an anchor, whose baseline definition is replaced by our banked construction later.

3.2. Energy-based contrastive objective

For an anchor–positive pair $(\mathbf{z}^{(a)}, \mathbf{z}^{(p)})$, we adopt the InfoNCE-style loss:

$$\ell(\mathbf{z}^{(a)}, \mathbf{z}^{(p)}) = -\log \frac{\exp\big(s(\mathbf{z}^{(a)}, \mathbf{z}^{(p)})/\tau\big)}{\exp\big(s(\mathbf{z}^{(a)}, \mathbf{z}^{(p)})/\tau\big) + \sum_{\mathbf{n}\in\mathcal{N}(\mathbf{z}^{(a)})} \exp\big(s(\mathbf{z}^{(a)}, \mathbf{n})/\tau\big)}. \tag{1}$$

The numerator enforces alignment between correlated views; the denominator spreads representations by penalizing high-similarity impostors, shaping the alignment–uniformity geometry (Wang & Isola, 2020) and admitting the mutual-information view of noise-contrastive estimation (Poole et al., 2019; van den Oord et al., 2018). In-batch negatives are efficient yet often uninformative or unstable, whereas a full softmax over all users/items is informative but costly and prone to false negatives in collaborative settings. We therefore preserve the baseline positives and logits and redefine $\mathcal{N}(\cdot)$ to be informative, fresh, and safe at a cost that does not scale with catalog size.

3.3. EMM: A plug-and-play memory-banked negative module.

The module touches only the negative set in Eq. (1). At step t we maintain clean-view momentum banks for users and items:

$$\mathbf{B}_u^{(t)} \leftarrow m\,\mathbf{B}_u^{(t-1)} + (1-m)\,\widetilde{\mathbf{Z}}_u^{(t)}, \qquad \mathbf{B}_i^{(t)} \leftarrow m\,\mathbf{B}_i^{(t-1)} + (1-m)\,\widetilde{\mathbf{Z}}_i^{(t)}, \tag{2}$$

where $m \in [0, 1)$ controls temporal smoothing and $\widetilde{\mathbf{Z}}^{(t)}$ are non-augmented embeddings. To avoid drift accumulation and synchronize with the evolving encoder, we periodically rebuild the banks by copying current clean embeddings,

$$\mathbf{B}_u^{(t)} \leftarrow \widetilde{\mathbf{Z}}_u^{(t)}, \qquad \mathbf{B}_i^{(t)} \leftarrow \widetilde{\mathbf{Z}}_i^{(t)}, \tag{3}$$

For scalable retrieval we form fixed-size candidate pools $C_u \subseteq \{1, \dots, M\}$ and $C_i \subseteq \{1, \dots, N\}$, and roll-refresh a small fraction of indices each step, keeping per-step cost independent of M, N while ensuring that candidates reflect the current geometry. If ANN search is used, its index is rebuilt alongside Eq. (3). Given an anchor $\mathbf{z}^{(a)}$ and a candidate matrix $\mathbf{C}$ whose rows are bank vectors addressed by C, we rank by a normalized similarity

$$\hat{s}\big(\mathbf{z}^{(a)}, \mathbf{c}\big) = \frac{\mathbf{z}^{(a)\top}\mathbf{c}}{\|\mathbf{z}^{(a)}\|\,\|\mathbf{c}\|}$$

for scale-robust ordering, select the Top-K hardest, and add an exploratory random quota $r = \lceil \alpha K \rceil$ from the same pool,

$$\mathcal{N}\big(\mathbf{z}^{(a)}\big) = \mathrm{TopK}\big(\mathbf{C}, \hat{s}, K\big) \,\cup\, \mathrm{Rand}\big(\mathbf{C}, r\big), \qquad \alpha \in [0.05,\, 0.2], \tag{4}$$

with ID exclusion to avoid leaking the true positive. A short warm-up with $K = 0$ stabilizes early dynamics before enabling hard mining. Crucially, we keep logits and positive construction identical to the baseline and replace only the denominator's negative sum. For LightGCL-style objectives that explicitly separate the positive term and the $\log \sum \exp$ part, we keep the positive term exact and approximate only the negative sum with $\mathcal{N}(\cdot)$. This preserves the geometry and scale of the original objective while delivering informative, fresh, and safe negatives at modest cost.

Design choices warrant additional remarks. Using clean embeddings in Eq. (2) makes the bank a longer-horizon summary of the representation space; injecting augmented noise would contaminate the notion of "hardness". The momentum m trades variance for lag; the rebuild period T resets lag and synchronizes any ANN index. Rolling refresh yields ergodicity: even with a fixed pool size, anchors eventually meet diverse candidates. Separating ranking $\hat{s}$ from scoring s preserves calibration of the denominator. Finally, scheduling K upward and α downward implements a natural curriculum from exploration to exploitation.

3.4. Why it works: Consistent approximation, gradient shaping, robustness

The module improves training dynamics because it supplies a denominator that is (i) a consistent surrogate of the full log–sum–exp, (ii) gradient-aligned with controlled variance, (iii) robust to false negatives, and (iv) stable over time via EMA and periodic rebuild.
Let

$$Z_{\mathrm{full}} = e^{u} + \sum_{j \in \Omega} e^{v_j}, \qquad Z_C = e^{u} + \sum_{j \in C} e^{v_j},$$

with $u = s(\mathbf{z}^{(a)}, \mathbf{z}^{(p)})/\tau$, $v_j = s(\mathbf{z}^{(a)}, \mathbf{n}_j)/\tau$, full index set Ω, and candidate set C from Eq. (4). The log-domain error obeys

$$0 \;\le\; \log Z_{\mathrm{full}} - \log Z_C = \log\!\Big(1 + \frac{\sum_{j \in \Omega \setminus C} e^{v_j}}{Z_C}\Big) \;\le\; \frac{\sum_{j \in \Omega \setminus C} e^{v_j}}{Z_C}, \tag{5}$$

where the last inequality uses $\log(1 + x) \le x$ for $x \ge 0$. Because $\log \sum \exp$ is exponentially dominated by a small head of large v_j, Top-K captures most probability mass, while the random-mix explicitly samples the tail, reducing systematic bias. As the pool grows and K approaches the pool size,

$$\log Z_C \;\longrightarrow\; \log Z_{\mathrm{full}}, \tag{6}$$

so optimizing Eq. (1) with $\mathcal{N}(\cdot)$ recovers the original InfoNCE in the limit, preserving its mutual-information character. The temperature τ modulates the spectrum: small τ sharpens the head making Top-K particularly effective but increases sensitivity to staleness, whereas large τ flattens it and raises the importance of the random share α for coverage.

Turning to gradients, write $\pi_p = e^{u}/Z$ and $\pi_j = e^{v_j}/Z$ with $Z \in \{Z_{\mathrm{full}}, Z_C\}$. For dot-product similarity,

$$\frac{\partial \ell}{\partial \mathbf{z}^{(a)}} = \frac{1}{\tau}\Big((\pi_p - 1)\,\mathbf{z}^{(p)} + \sum_j \pi_j\, \mathbf{n}_j \Big). \tag{7}$$

Negatives with large π_j dominate the gradient; Top-K therefore allocates computation where it matters most, instead of relying on incidental in-batch composition. Interpreting the negative sum as importance sampling with proposal q induced by Top-K + mix yields the variance sketch

$$\mathrm{Var}\big[\nabla \ell\big] \;\approx\; \mathrm{Var}_{\,j \sim q}\!\left[\frac{\pi_j}{q_j}\,\mathbf{g}_j\right], \qquad \mathbf{g}_j = \frac{\mathbf{n}_j}{\tau}, \tag{8}$$

so concentrating q on high π_j reduces variance by keeping π_j/q_j nearly constant over the head, while the random share caps the tail and prevents brittleness. A brief warm-up with K=0 avoids committing to unreliable "hardness" early; ramping K and annealing α implements a robust curriculum.

False negatives—semantically close nodes mislabeled as negatives—are particularly harmful under small τ. Two safeguards mitigate this: ID exclusion forbids the paired positive from entering $\mathcal{N}(\cdot)$, and a similarity window removes candidates with $\mathrm{sim} \ge \delta_{\mathrm{hi}}$ (likely positives) or $\mathrm{sim} \le \delta_{\mathrm{lo}}$ (uninformative). If the raw false-negative rate is ρ, naïve InfoNCE incurs $O(\rho)$ bias in expectation; filtering reduces it to

$$O\big(\rho \cdot \Pr\{\mathrm{sim} \le \delta_{\mathrm{hi}}\}\big), \tag{9}$$

attenuating over-repulsion without sacrificing hardness where it is safe.

Finally, stability arises from EMA tracking and periodic rebuild. Unrolling Eq. (2) gives

$$\mathbf{B}^{(t)} = m^{t}\mathbf{B}^{(0)} + (1 - m)\sum_{k=1}^{t} m^{t-k}\,\widetilde{\mathbf{Z}}^{(k)}, \tag{10}$$

and if smooth evolution $\big\|\widetilde{\mathbf{Z}}^{(k)} - \widetilde{\mathbf{Z}}^{(k-1)}\big\| \le L$ is applied, the tracking error is bounded by

$$\big\|\mathbf{B}^{(t)} - \widetilde{\mathbf{Z}}^{(t)}\big\| \;\le\; \frac{1 - m}{m}\,L. \tag{11}$$

The rebuild in Eq. (3) resets this error to zero at cadence T. Rolling refresh introduces ergodicity, so anchors do not repeatedly see the same few hard negatives; synchronizing any ANN index at rebuild points avoids search mismatch. Together these mechanisms eliminate the "stale hardest-only" failure mode and keep the denominator informative and well-behaved throughout training (Wu et al., 2018).

3.5. Computational complexity

Once the Top-K plus random mix is known, building logits and evaluating the contrastive loss costs

$$\mathcal{O}\big(B\,d\,(K{+}r)\big), \tag{12}$$

with batch size B, embedding dimension d, hard negatives K, and random quota r. This term is independent of catalog sizes M and N.

Identifying the Top-K inside a fixed candidate pool of size C adds a retrieval cost. With an exact scan:

$$\mathcal{O}(B\,d\,C) + \mathcal{O}(B\,C), \tag{13}$$

which scales linearly in C and remains independent of M, N. With an approximate index, query time is sublinear in C and re-scoring touches only $K{+}r$ candidates, so the runtime is close to Eq. (13). Bank maintenance is lightweight. EMA updates touch only rows seen at the current step, and periodic rebuild every T steps contributes an amortized overhead:

$$\mathcal{O}\!\left(\frac{(M{+}N)\,d}{T}\right), \tag{14}$$

Rolling refresh replaces only a small fraction of pool entries per step, adding a minor bandwidth-bound cost. In practice, the dominant operation is a single batched similarity kernel or index probing; because

Table 1
Statistics of datasets.

Dataset	# Users	# Items	# Interactions	Sparsity
Baby	19,445	7,050	160,792	99.88%
Sports	35,598	18,357	296,337	99.95%
Clothing	39,387	23,033	237,488	99.97%

C and K are fixed and modest, end-to-end cost is predictable and essentially independent of M, N, while yielding stronger and more stable gradients than in-batch negatives at a fraction of full-softmax compute.

4. Experiments

4.1. Experimental settings

Datasets. The dataset statistics used in this study are summarized in Table 1. We primarily evaluate on three widely adopted Amazon category subsets: *Baby*, *Clothing*, and *Sports*, which are constructed from the public Amazon product review corpus (Ni et al., 2019). Each corpus is treated as an implicit-feedback bipartite graph of users and items, where an observed interaction denotes positive feedback and all unobserved pairs are considered candidates at evaluation time. To ensure comparability, we follow the released SSLRec splits and preprocessing protocol (Ren et al., 2024). No textual or visual side information is used in our main experiments; all methods operate purely on the interaction graph derived from implicit feedback.

Metrics. We evaluate recommendation quality with Top-K ranking measures that emphasize accuracy at the head of the list, where user utility concentrates (Cremonesi et al., 2010). In all experiments, the model ranks the full catalog for each user after excluding that user's training interactions (i.e., no sampled negatives at test). Metrics are computed per user and then macro-averaged over the test set; ties are deterministically broken by model score. We report Recall@K and NDCG@K for $K \in \{10, 20\}$, matching the columns in our result tables.

- **Recall@K** measures whether the ground-truth items are covered within the Top-K, thus reflecting coverage of relevant targets under the leave-one-out protocol.
- **NDCG@K** measures rank-sensitive utility by discounting lower-ranked hits and normalizing by the ideal list, thereby rewarding correct ordering at early ranks (Järvelin & Kekäläinen, 2002).

Unless otherwise specified, model selection is performed by validation NDCG@20, and we report Recall@10/20 and NDCG@10/20 on the test set as our primary endpoints. All metrics are normalized to $[0, 1]$, with higher being better.

Baselines. We compare against five representative graph-based, contrastive recommenders and report each backbone in its original form as well as with our plug-in attached. We additionally include GNS as a plug-in baseline that follows an IRGAN-style generative-negatives paradigm for negative construction (Wang et al., 2017). The plug-in operates only at the sampling/loss layer and does not modify the forward propagation of any baseline.

SGL (Wu et al., 2021). A self-supervised graph learning framework that constructs multiple stochastic graph-dropout views on the user–item bipartite graph and optimizes an InfoNCE objective to align view-consistent representations; it is among the earliest and most widely adopted CL backbones for recommendation.

SimGCL (Yu et al., 2022). A simplified contrastive learner that discards heavy graph augmentations and instead injects lightweight noise in embedding space to generate two views, thereby stabilizing training while preserving collaborative semantics; it is a strong, minimal baseline for plug-in evaluation.

LightGCL (Cai et al., 2023). A globally regularized contrastive paradigm that leverages SVD-based refinement to form semantics-preserving views without stochastic edge/node drop, improving robustness to augmentation noise and achieving state-of-the-art performance across sparse regimes.

NCL (Lin et al., 2022). Neighborhood-enriched contrastive learning, which explicitly incorporates potential neighbor relations into contrastive pair construction so that local collaborative structure is emphasized during representation learning.

DCCF (Ren et al., 2023). Disentangled contrastive collaborative filtering, which introduces intent disentanglement and adaptive interaction augmentation to separate latent factors while supervising them with a contrastive signal, yielding stronger generalization and robustness.

Implementation Details. We implement all models in PyTorch within SSLRec and train/evaluate on a single NVIDIA RTX 4070 GPU, keeping each backbone's released settings intact. Our method is attached only at the sampling/loss layer; forward propagation and supervised objectives remain unchanged. Unless stated otherwise, we use Adam (Kingma & Ba, 2015) with the backbone's defaults and apply early stopping by validation NDCG@20. The plug-in adds a lightweight dynamic memory and a mining policy that retrieves Top-K hard negatives with a small random-negative mix and a short warm-up. Plug-in hyperparameters are tuned on the validation set while holding all backbone settings fixed. At test time we rank the full catalog per user, compute metrics per user, and macro-average; checkpoints are selected by validation NDCG@20, and we report Recall@10/20 and NDCG@10/20.

4.2. Overall performance

As shown in Table 2, we evaluate EMM (Episodic-Memory Mining) on five backbones and three benchmarks while keeping baseline positives and forward propagation unchanged, so that any gains primarily reflect negative-sampling fragility being addressed. We additionally report an IRGAN-style generative-negatives plug-in (GNS) under the same attachment protocol to strengthen the comparison on negative construction

Across five representative backbones (SGL, SimGCL, LightGCL, NCL, DCCF) and three benchmarks (Baby, Clothing, and Sports), attaching our plug-in EMM yields uniform improvements in both Recall@K and NDCG@K., and GNS is consistently competitive over the same backbones and datasets.

The effect is architecture-agnostic: methods based on stochastic graph dropout, lightweight embedding perturbations, SVD-guided refinement, neighborhood contrast, and dual-channel collaborative filtering all benefit, which indicates that principled negative sample construction complements existing view designs rather than competing with them. Notably, view-heavy backbones (e.g., SGL, SimGCL) also improve without altering positives, suggesting EMM does not amplify augmentation noise. Overall, the average improvement in ranking quality is around a consistent double-digit range across benchmarks at comparable training cost. Improvements are mirrored at cutoffs 10 and 20, which shows that EMM enhances not only the probability of a hit but also the quality of early-rank ordering. Together with our Layer-number sensitivity (Sec. 4.4), the gains degrade more slowly with depth, indicating reduced sensitivity to over-smoothing even though the message-passing architecture is unchanged. Compared with GNS, EMM achieves the best or tied-best results in the large majority of cases, while GNS occasionally matches or slightly exceeds EMM on a few metrics, indicating that EMM provides a more stable improvement overall.

Table 2
Top-K recommendation performance of baselines and their variants with GNS/EMM on Baby, Clothing, and Sports. R@10/20 and N@10/20 denote Recall@10/20 and NDCG@10/20. All results are averaged over five independent runs with different random seeds. Statistical significance is assessed by paired two-sided t-tests, and the reported improvements are significant with $p < .05$ (marked by †). Best results are in **bold** and second-best are underlined within each model group.

Model	Baby				Clothing				Sports			
	R@10	R@20	N@10	N@20	R@10	R@20	N@10	N@20	R@10	R@20	N@10	N@20
SGL	0.0368	0.0558	0.0197	0.0245	0.0284	0.0417	0.0153	0.0186	0.0501	0.0734	0.0268	0.0327
SGL w/GNS	0.0410	**0.0624**	0.0221	0.0275	0.0304	0.0450	0.0165	0.0202	0.0539	0.0795	0.0289	0.0353
SGL w/EMM	**0.0426**	**0.0624**	**0.0228**	**0.0278**	**0.0362**	**0.0528**	**0.0200**	**0.0242**	**0.0589**	**0.0866**	**0.0316**	**0.0386**
Improv.	15.59%	11.79%	15.82%	13.38%	27.45%	26.62%	31.32%	29.97%	17.60%	17.93%	17.88%	17.98%
SimGCL	0.0445	0.0703	0.0234	0.0297	0.0274	0.0415	0.0147	0.0182	0.0524	0.0797	0.0277	0.0346
SimGCL w/GNS	0.0464	0.0730	0.0244	0.0310	0.0282	0.0424	0.0150	0.0187	0.0543	**0.0828**	0.0288	0.0359
SimGCL w/EMM	**0.0475**	**0.0746**	**0.0251**	**0.0319**	**0.0296**	**0.0449**	**0.0159**	**0.0197**	**0.0547**	0.0826	**0.0290**	**0.0361**
Improv.	6.74%	6.18%	7.08%	7.26%	7.88%	8.09%	8.02%	8.22%	4.51%	3.66%	4.76%	4.28%
LightGCL	0.0446	0.0664	0.0242	0.0297	0.0346	0.0492	0.0185	0.0223	0.0574	0.0846	0.0311	0.0379
LightGCL w/GNS	0.0465	0.0722	0.0248	0.0312	0.0370	0.0516	0.0197	0.0236	0.0587	0.0861	0.0318	0.0387
LightGCL w/EMM	**0.0499**	**0.0766**	**0.0266**	**0.0333**	**0.0379**	**0.0564**	**0.0205**	**0.0252**	**0.0597**	**0.0905**	**0.0320**	**0.0395**
Improv.	11.78%	15.30%	10.26%	12.33%	9.60%	14.46%	11.27%	12.81%	4.11%	6.88%	2.90%	4.27%
NCL	0.0426	0.0665	0.0224	0.0284	0.0177	0.0268	0.0098	0.0120	0.0447	0.0674	0.0239	0.0296
NCL w/GNS	0.0462	0.0723	0.0242	0.0307	0.0209	0.0310	0.0114	0.0140	0.0531	0.0804	0.0284	0.0353
NCL w/EMM	**0.0479**	**0.0750**	**0.0250**	**0.0318**	**0.0244**	**0.0363**	**0.0132**	**0.0162**	**0.0544**	**0.0816**	**0.0286**	**0.0355**
Improv.	12.29%	12.65%	11.43%	11.89%	37.43%	35.17%	35.25%	34.61%	21.60%	20.97%	19.87%	19.95%
DCCF	0.0338	0.0515	0.0182	0.0226	0.0319	0.0471	0.0173	0.0212	0.0538	0.0782	0.0288	0.0350
DCCF w/GNS	0.0378	0.0597	0.0200	0.0255	0.0337	0.0503	0.0183	0.0223	0.0578	**0.0869**	**0.0305**	**0.0378**
DCCF w/EMM	**0.0433**	**0.0683**	**0.0229**	**0.0291**	**0.0362**	**0.0522**	**0.0197**	**0.0237**	**0.0585**	0.0868	0.0303	0.0375
Improv.(EMM)	28.22%	32.54%	25.71%	28.62%	13.54%	10.78%	13.38%	11.80%	8.74%	10.99%	5.27%	7.03%
Avg.Improv. †	14.92%	15.69%	14.06%	14.70%	19.18%	19.02%	19.85%	19.48%	11.31%	12.09%	10.13%	10.70%

Equally important, a recurrent pattern emerges in harder and sparser regimes. On the sparsest datasets the relative gains are most pronounced, while in denser settings they remain positive albeit smaller. This observation aligns with the intended mechanism: a refreshable EMA memory supplies a stable yet up-to-date candidate distribution, and warm-started Top-K mining with a small random mix sharpens gradients without inflating the rate of false negatives. In sparse data, where in-batch negatives are often easy or stale, this combination reduces estimator variance and provides hard-but-safe contrasts, which translates into the observed improvements. EMM delivers these benefits without altering forward propagation or supervised losses, and with only modest computational overhead due to a fixed memory size and periodic rebuild, making it a practical drop-in enhancement for a broad class of contrastive recommenders. In summary, EMM delivers a consistent and meaningful uplift in ranking quality across all evaluated backbones and datasets. The results primarily address negative-sampling fragility, do not exacerbate augmentation noise despite leaving positives unchanged, and mitigate depth sensitivity observed under deeper propagation. The improvements appear reliably at both cutoffs and translate into sharper early ordering, stronger coverage of relevant items, and greater stability under sparse supervision. Taken together, these results establish EMM as a practical and effective plug-in that raises the performance ceiling of mainstream contrastive recommenders without architectural modification.

4.3. Ablation studies

Ablation on Memory vs. Mining. We decompose the plug-in into two functional parts and test them on DCCF and LightGCL over *Baby*, *Clothing*, and *Sports* while keeping all backbone settings fixed. The first part, referred to as the *Memory*, maintains a refreshed pool of user and item representations through exponential moving average (EMA) updates, periodic rebuild, and light rolling refresh. The second part, termed *Mining*, retrieves hard negatives from that pool by similarity, mixes a small fraction of random negatives, and applies a short warm-up phase using random sampling only. For each backbone we report four variants in Table 3: the released baseline, baseline with the memory module only, baseline with the mining module only, and the combined variant that enables both modules. We have the following observations:

- **Each module alone is effective and complementary.** Introducing only the memory module improves ranking quality by supplying a stable yet up-to-date candidate distribution, which reduces estimator variance and avoids stale or trivially easy negatives. Introducing only the mining module yields comparable gains by concentrating gradient signal on informative contrasts, while the random mix and warm-up restrain false negatives. As a representative example, DCCF on *Baby* gains roughly 25% in Recall@20 with the memory module alone, and LightGCL on *Baby* gains about 14% in the same metric with the memory module alone.
- **The combined variant is consistently superior.** Enabling both modules delivers the strongest results across datasets and backbones. The combined effect surpasses either single module because the refreshed distribution calibrates *what* is sampled while the mining policy shapes *how* it is used. On DCCF with *Baby*, Recall@20 rises by about 29% and NDCG@20 by more than 23%, illustrating the additive nature of the two parts.
- **Benefits amplify under sparsity while remaining stable elsewhere.** Gains are most pronounced on the sparser corpora and remain positive on the denser one. This pattern accords with the design intention. In sparse regimes, in-batch negatives tend to be easy or outdated; the refreshed memory addresses freshness and diversity, and the mining policy supplies hard but safe contrasts without destabilizing training. The result is sharper early ordering and improved coverage without modifying forward propagation or supervised losses.
- **Overhead is modest and orthogonal to architecture.** Both modules add little computational cost because the memory size is fixed and rebuilds are periodic, and neither alters message passing or inference. The ablation therefore isolates sample construction on the

Table 3
Ablation study on Baby, Clothing, and Sports. We decouple the plug-in into a refreshed memory (**Memory**) and a mining policy (**Mining**). +EMM enables both. All backbone settings are kept fixed.

Model	Baby				Clothing				Sports			
	R@10	R@20	N@10	N@20	R@10	R@20	N@10	N@20	R@10	R@20	N@10	N@20
DCCF	0.0346	0.0526	0.0189	0.0234	0.0323	0.0460	0.0179	0.0213	0.0548	0.0807	0.0292	0.0357
DCCF w/Memory	0.0401	0.0655	0.0214	0.0279	0.0345	0.0508	0.0182	0.0223	0.0578	0.0880	0.0304	0.0380
DCCF w/Mining	0.0398	0.0654	0.0214	0.0278	0.0355	0.0515	0.0184	0.0226	0.0576	0.0872	0.0303	0.0379
DCCF w/EMM	0.0433	0.0676	0.0228	0.0289	0.0351	0.0523	0.0188	0.0232	0.0601	0.0902	0.0316	0.0392
LightGCL	0.0430	0.0641	0.0234	0.0287	0.0322	0.0475	0.0177	0.0215	0.0568	0.0828	0.0302	0.0367
LightGCL w/Memory	0.0480	0.0729	0.0253	0.0316	0.0353	0.0514	0.0194	0.0235	0.0581	0.0871	0.0306	0.0381
LightGCL w/Mining	0.0479	0.0725	0.0254	0.0316	0.0350	0.0515	0.0194	0.0236	0.0579	0.0872	0.0305	0.0379
LightGCL w/EMM	0.0513	0.0789	0.0266	0.0335	0.0368	0.0544	0.0202	0.0246	0.0589	0.0898	0.0310	0.0386

negative side as the driver of improvement rather than extra capacity or heavier encoders.

4.4. Sensitivity analysis

We now conduct a sensitivity analysis of the main hyperparameters and the EMM plug-in. First, we examine the effect of the temperature τ using three representative backbone–dataset pairs: DCCF on *Baby*, LightGCL on *Sports*, and NCL on *Clothing*. Next, we study embedding size with DCCF on all three datasets. We then vary the layer number for DCCF on the same datasets. Finally, we probe EMM's internal knobs, candidate_size and hard_k, using DCCF and LightGCL on *Baby* to isolate sampling effects.

We additionally study the mixing ratio α that trades off hard negatives from the candidate pool and uniformly sampled random negatives, and evaluate its robustness across datasets and sparsity-controlled settings.

To further examine the reliability of hard-negative mining, we ablate two lightweight false-negative safeguards, namely ID exclusion and a cosine-similarity window, and report the sensitivity of the similarity-window thresholding with a coarse sweep.

Finally, we provide a fine-grained runtime breakdown that decomposes EMM overhead into the ANN pipeline term and the memory-bank maintenance term, normalized against baseline sampling, to clarify where the additional cost originates. Unless otherwise noted, we vary one factor at a time while keeping the remaining settings at their default values, and we report standard Top-K ranking metrics.

4.4.1. Effect of temperature τ

We examine how the temperature controls the hardness of negatives in InfoNCE and how it interacts with EMM. Fig. 2 plots performance across τ for three representative pairs: DCCF on *Baby*, LightGCL on *Sports*, and NCL on *Clothing*. Three consistent observations emerge:

- Moving from a very small τ to a moderate value improves discrimination among negatives. With EMM, this improvement is more pronounced and the high-performance region becomes wider, indicating reduced sensitivity to the exact choice of τ.
- An overly large τ softens contrast and weakens the model's ability to separate hard from easy negatives, which degrades performance. EMM alleviates this degradation by supplying informative candidates and filtering false negatives, though it does not remove the effect entirely.
- Compared with the baselines, EMM tends to shift the preferred τ slightly upward and stabilizes optimization, leading to smoother training curves and a broader plateau of strong results.

In practice, we recommend tuning τ in a moderate range. Very small values concentrate gradients on a few negatives and may destabilize training, while very large values blur hardness. EMM makes the method less brittle with respect to τ by combining momentum-updated positives, candidate-pool mining, and false-negative suppression.

4.4.2. Effect of embedding size d

We study how the embedding dimension balances representation capacity and optimization stability under EMM, using DCCF on *Baby*, *Clothing*, and *Sports*. Three consistent patterns emerge; results are shown in Fig. 3.

- Increasing d from a small to a moderate level improves discrimination and yields smoother validation curves, indicating better expressiveness without sacrificing stability. This trend appears on all three datasets.
- Pushing d excessively high brings diminishing returns and slows convergence, and the validation trajectory may fluctuate more before early stopping. This is most visible on the larger and sparser settings, where over-parameterization makes optimization less efficient.
- With EMM, the high-performance region across d becomes broader and less brittle, since the memory bank, candidate-pool mining, and false-negative suppression provide informative contrast even when the representation is not maximally wide. In practice, a moderate d offers a good trade-off and can be fixed for subsequent ablations to reduce tuning burden.

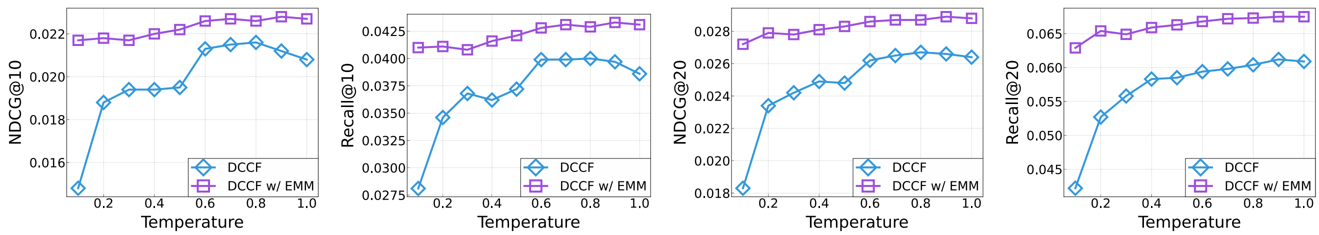
4.4.3. Effect of layer number

We study how graph depth affects representation mixing and optimization under EMM by varying the layer number of DCCF on *Baby*, *Clothing*, and *Sports*. Fig. 4 plots performance across depths with consistent patterns on all three datasets:

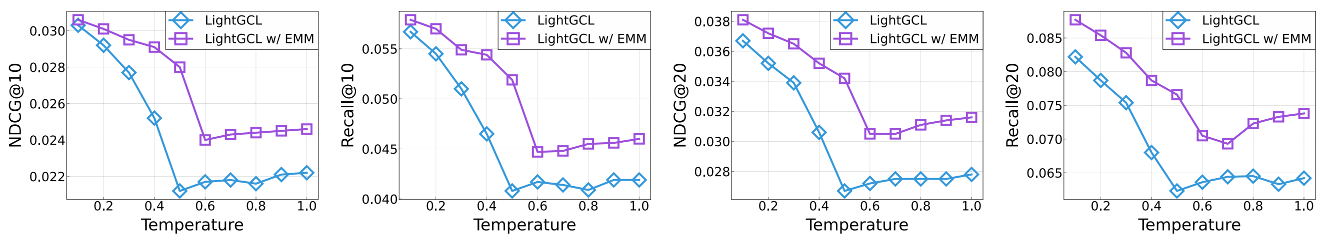
- Shallow to moderate depth is most effective: Baby peaks at one layer, while Clothing and Sports peak at two. A modestly larger receptive field supplies the memory with structure aware keys, so Top K mining finds stronger negatives and the random mix preserves diversity. Around these depths, validation curves are smooth and the high quality region is broad, especially on Clothing and Sports.
- Increasing depth further degrades ranking quality. With three to four layers the representation changes faster than the EMA memory can track, creating a mismatch between queries and candidates and raising the chance of borderline false negatives. Periodic rebuilds and similarity window filtering mitigate the issue, but returns diminish and training becomes slower and more variable.
- A practical default emerges. Two layers work reliably across datasets, while one layer remains competitive on Baby where interactions are scarcer and local structure dominates. EMM maintains freshness, diversity, and safety of negatives across these settings, which reduces sensitivity to the exact depth and allows tuning to focus on other hyperparameters.

4.4.4. Effect of candidate_size and hard_k

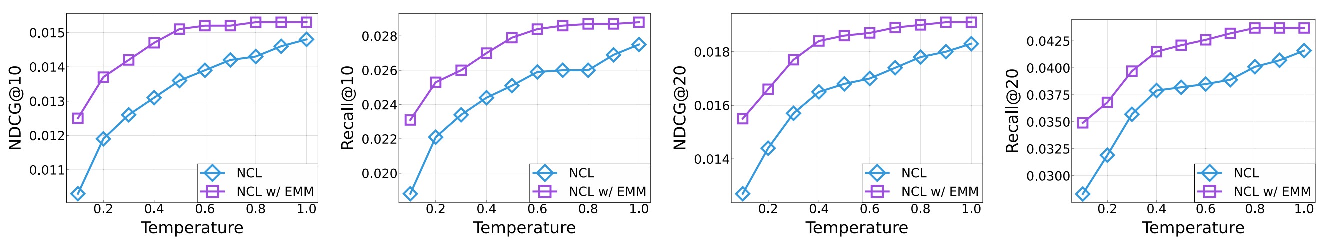
We visualize the joint effect of the sampler size and the number of mined negatives with heatmaps on *Baby*, using DCCF (w/EMM) and LightGCL (w/EMM). As illustrated in Fig. 5, three consistent patterns emerge across both backbones.



(a) DCCF on Baby



(b) LightGCL on Sports



(c) NCL on Clothing

Fig. 2. EMM performance as adjusting τ.

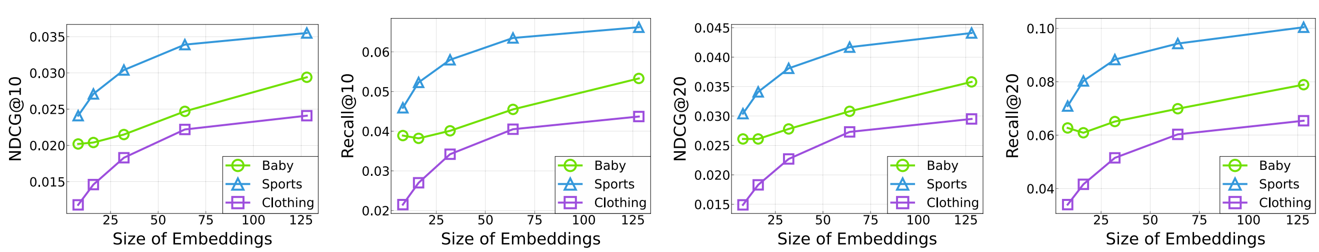


Fig. 3. EMM performance as adjusting d.

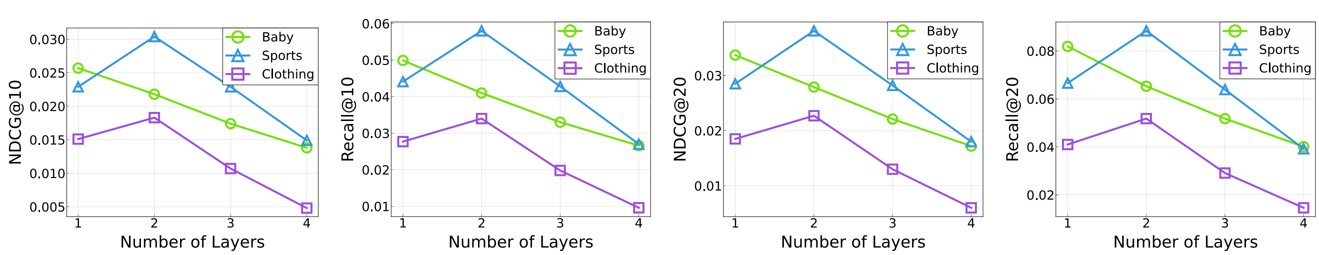
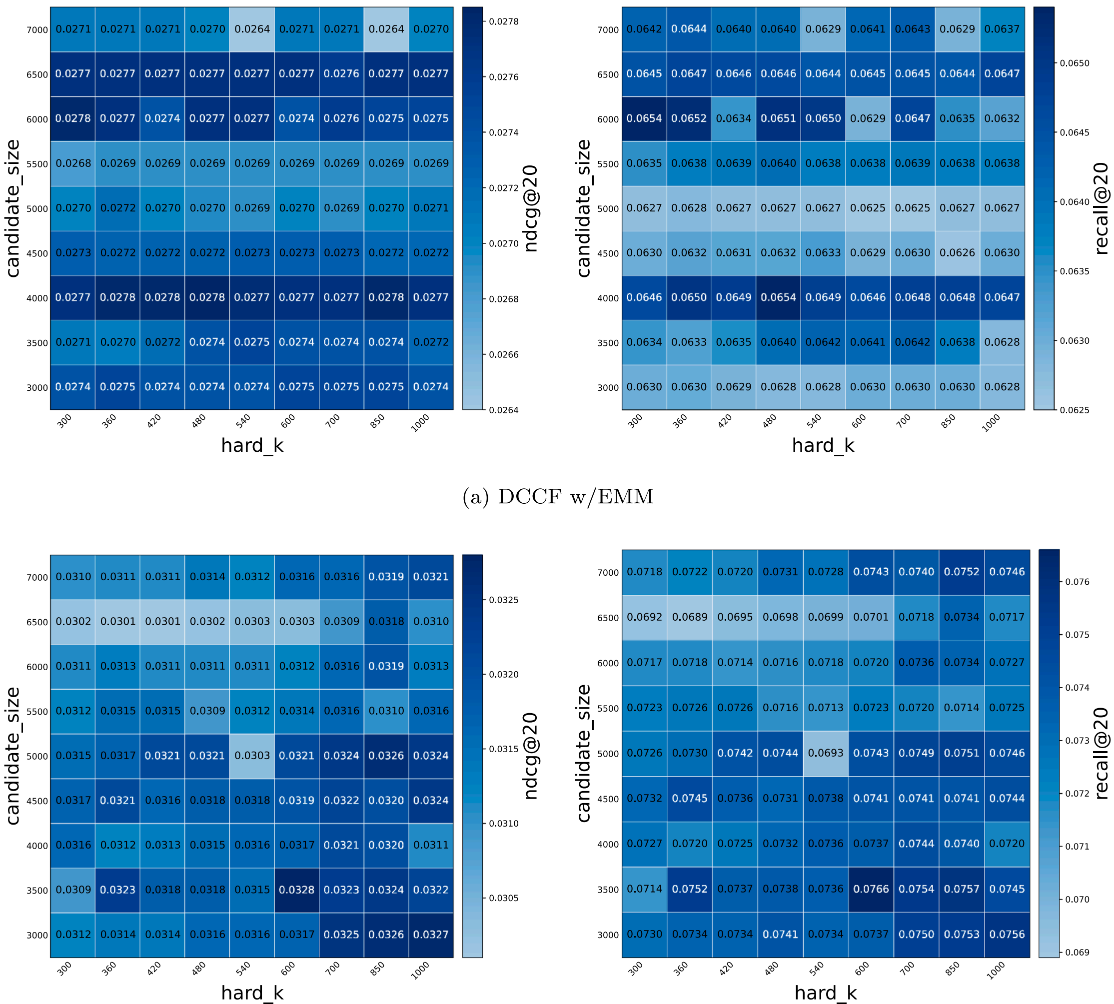


Fig. 4. EMM performance as adjusting layer numbers.

(1) Very small candidate_size limits diversity in the candidate pool and produces a low-performance basin. Enlarging the pool lifts performance until the pool becomes sufficiently diverse; further growth shows diminishing returns.
(2) For a fixed pool size, increasing hard_k from a small value first helps by bringing in informative negatives, then hurts once selection becomes too aggressive. Excessively large hard_k injects noisy or borderline false negatives that destabilize training. The sweet region appears at moderate hard_k.
(3) The high-performance area forms a ridge that runs along moderate `candidate_size` with sublinear growth in `hard_k`. EMM widens this ridge by filtering likely false negatives and mixing batch keys with memory-bank keys, which reduces sensitivity to the exact pair of settings. LightGCL shows slightly sharper sensitivity to `hard_k` than



(a) DCCF w/EMM

(b) LightGCL w/EMM

Fig. 5. Heatmaps on *Baby*.

DCCF, while both maintain a broad plateau once the pool is sufficiently large.

Practical recipe. Start from a moderate candidate_size that comfortably exceeds the batch and set hard_k to grow sublinearly with the pool. If the heatmap shows a flat ridge, prefer the left side of that ridge to reduce computation while keeping accuracy stable. Once a stable pair is chosen, fix them for the remaining ablations and tune other factors.

4.4.5. Robustness of EMM under data sparsity

To examine robustness to user–item sparsity, we partition users into groups by interaction counts (0–5, 5–10, 10–15, 15–20) and report performance for three representative backbone–dataset pairs: DCCF on Baby, SGL on Sports, and NCL on Clothing. The results are shown in Fig. 6.

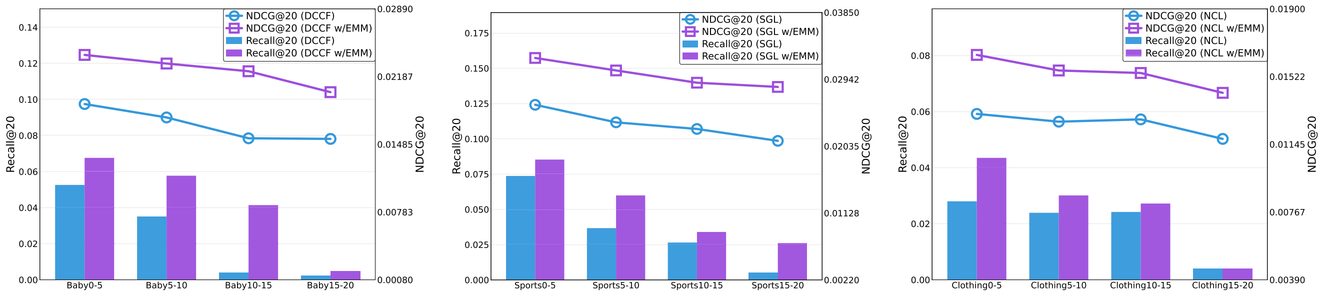
Across all groups and all pairs, attaching EMM yields consistent lifts in both Recall@K and NDCG@K. The gains are most pronounced in the low-activity regimes, where in-batch negatives tend to be either stale or trivially easy; EMM supplies a refreshed candidate distribution and focuses learning on informative contrasts, thereby stabilizing the optimization. As interaction increases, improvements remain positive and the curves become smoother, indicating reduced sensitivity to the exact sparsity level.

Comparing the baselines, we observe larger fluctuations across sparsity groups, especially for the contrastive-only variants that rely on incidental batch composition. With EMM, these fluctuations are attenuated: the EMA-updated memory and dynamic hard-negative mining maintain diversity and freshness while suppressing likely false negatives, which preserves early-rank quality even for the hardest user groups. Overall, the results support that principled negative construction is an effective lever for alleviating the adverse effects of sparsity without modifying backbone forward propagation.

4.4.6. Sensitivity study on α.

Our sampler mixes two types of negatives: (i) hard negatives retrieved from the candidate pool and (ii) random negatives uniformly sampled from the item universe. Their mixture is controlled by α, where larger α injects more random negatives. Hard negatives improve discriminative learning but may increase the chance of false negatives, while random negatives promote diversity and stabilize optimization yet can become less informative. Hence, α governs a trade-off between hardness and diversity.

We examine the sensitivity to $\alpha \in \{0.05, 0.10, 0.15, 0.20, 0.40, 1.00\}$ with all other hyperparameters fixed, and report Recall@20 on Baby, Clothing, and Sports, each further evaluated under multiple sparsity-controlled variants . Across these settings, performance consistently peaks in a narrow low-α region, with the best Recall@20 typically achieved at $\alpha = 0.15$ or 0.20 , forming a stable high-performing plateau over $\alpha \in [0.05, 0.20]$. In contrast, when α increases beyond this region, Recall@20 exhibits a clear degradation trend: $\alpha = 0.40$ already leads to noticeable drops, and $\alpha = 1.00$ further amplifies the decline.



(a) DCCF on Baby (b) SGL on Sports (c) NCL on Clothing

Fig. 6. EMM performance on sparse datasets. Users are grouped by interaction counts (0–5, 5–10, 10–15, 15–20).

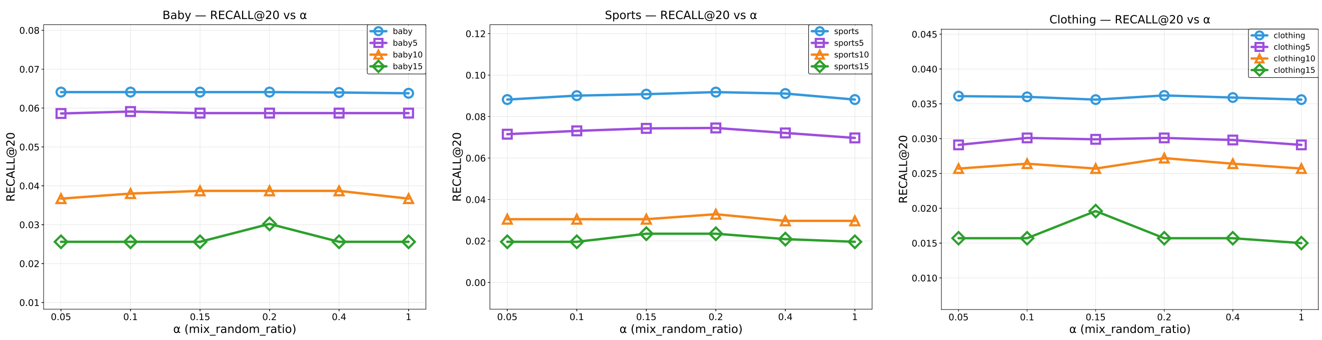


Fig. 7. Sensitivity of the random-mix ratio α under different sparsity settings.

This pattern is consistent with the underlying trade-off: overly large α over-injects random negatives, which dilutes hard-negative supervision and weakens the training signal. Overall, the results indicate that the recommended range $\alpha \in [0.05, 0.20]$ remains reliable even under substantially different sparsity levels, and we therefore use $\alpha = 0.15$ as the default choice in the remaining experiments, as it lies in the consistently strong region across datasets and sparsity conditions. As shown in Fig. 7, the low-α regime consistently yields strong performance across datasets and sparsity conditions.

4.4.7. False-negative filtering via ID exclusion and similarity window.

Hard-negative retrieval may introduce false negatives that are semantically close to the anchor but are treated as negatives, leading to over-repulsion in contrastive learning. We mitigate this issue with two lightweight safeguards: *ID exclusion* and a *similarity window*. ID exclusion prevents the paired positive from entering the retrieved negative set. The similarity window filters candidates by cosine similarity between ℓ_2-normalized embeddings and retains only "safe" negatives:

$$\delta_{\mathrm{lo}} < \mathrm{sim}(\mathbf{z}, \mathbf{z}_j) < \delta_{\mathrm{hi}}, \tag{15}$$

where δ_{hi} removes overly similar candidates and δ_{lo} discards extremely dissimilar, typically uninformative negatives.

Threshold calibration. We fix a mild lower bound $\delta_{\mathrm{lo}} = -0.5$ so that suppression is only activated for potentially confusable negatives with non-trivial cosine similarity, while clearly irrelevant negatives with very low similarity are preserved to maintain negative diversity and optimization stability. With δ_{lo} fixed, we tune δ_{hi} on the validation set via a coarse 1D sweep. The sensitivity curves under $\delta_{\mathrm{lo}} = -0.5$ in Fig. 8 show a broad plateau around the selected δ_{hi}, indicating low tuning burden.

Ablation. Table 4 reports a 2×2 ablation of the two safeguards across representative backbone–dataset pairs. The similarity window consistently yields the main gains, while ID exclusion alone is typically marginal but non-degrading; enabling both achieves the best or tied-best results, supporting their complementary roles in suppressing false negatives from hard-negative retrieval.

Table 4
Ablation of false-negative safeguards: ID exclusion (ID ex.) and similarity window (sim w.). Best results are in **bold** and second-best are underlined within each dataset block.

Dataset	Model	R@10	R@20	N@10	N@20
Baby	DCCF	0.0404	0.0643	0.0216	0.0276
	w/ID ex.	0.0403	0.0643	0.0215	0.0275
	w/sim w.	0.0420	**0.0683**	0.0223	**0.0289**
	w/Both	**0.0423**	0.0682	**0.0224**	**0.0289**
Clothing	NCL	0.0229	0.0345	0.0124	0.0153
	w/ID ex.	0.0232	0.0349	0.0125	0.0154
	w/sim w.	0.0243	0.0366	0.0131	0.0162
	w/Both	**0.0247**	**0.0371**	**0.0133**	**0.0164**
Sports	LightGCL	0.0577	0.0874	0.0305	0.0379
	w/ID ex.	0.0576	0.0876	0.0306	0.0381
	w/sim w.	0.0602	0.0907	**0.0321**	0.0397
	w/Both	**0.0608**	**0.0912**	**0.0321**	**0.0398**

4.4.8. Runtime breakdown of EMM overhead.

We further provide a fine-grained runtime decomposition to clarify where EMM's computational overhead originates. All timings are measured as wall-clock training time per iteration in milliseconds under identical hardware and hyperparameters. We define *Base* as the per-iteration time of the baseline sampler and *EMM* as the per-iteration time with EMM enabled. We then attribute the additional EMM cost to two components. The first component, t_{over}_ann, captures the ANN pipeline overhead, including hard-negative retrieval and the amortized cost of index maintenance and rebuild. The second component, t_{over}_bank, captures the memory-bank maintenance overhead, including the amortized cost of bank refresh and rebuild and any required synchronization. We report the normalized overheads relative to baseline sampling.

As summarized in Table 5, the ANN component explains most of the additional runtime, whereas the bank component remains consistently small, indicating that EMM's overhead is primarily driven by ANN operations rather than memory-bank rebuild.

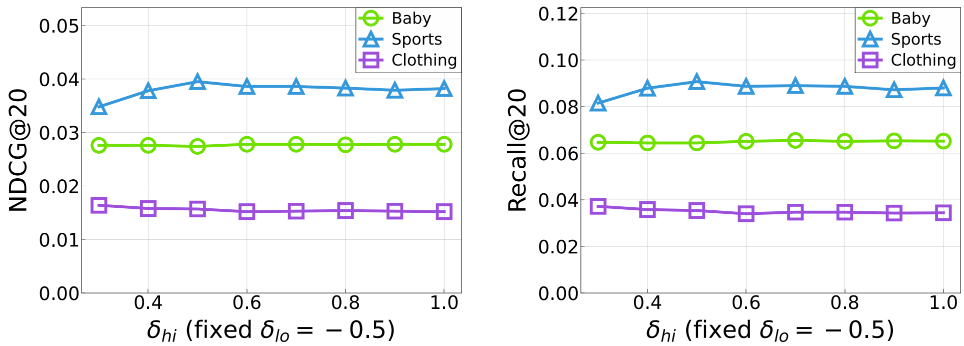


Fig. 8. Sensitivity curves for tuning δ_{hi} under fixed $\delta_{\text{lo}} = -0.5$ (left) and an additional analysis related to false-negative filtering (right).

Table 5
Runtime breakdown of EMM relative to baseline sampling. Values are mean training time per iteration in ms/iter. ANN% and Bank% are computed as $100 \times t_{\text{over}}$_ann/*Base* and $100 \times t_{\text{over}}$_bank/*Base*. t_{over}_ann covers ANN retrieval and amortized index maintenance, and t_{over}_bank covers amortized bank refresh.

	Baby				Sport				Clothing			
Model	Base	EMM	ANN%	Bank%	Base	EMM	ANN%	Bank%	Base	EMM	ANN%	Bank%
DCCF	39.0	59.9	106.6	1.6	58.1	78.8	90.0	1.2	62.6	90.8	99.6	1.1
NCL	41.3	75.0	19.7	0.2	55.3	81.4	15.4	0.2	66.2	101.3	12.8	0.1
LightGCL	5.4	7.4	5.6	1.9	9.3	9.8	3.2	1.1	10.1	9.7	3.0	1.0
SimGCL	29.9	50.4	93.9	0.6	49.2	66.6	57.4	0.4	54.8	70.7	51.2	0.2
SGL	29.0	65.6	87.0	0.3	49.9	118.7	93.1	0.4	56.4	133.5	93.3	0.3

5. Conclusion

We presented a plug-and-play module for contrastive recommendation that operates entirely at the sampling and loss layer, combining an exponential moving average (EMA) candidate memory with Top-K hard-negative mining plus a light random mix and a short warm-up. This design supplies hard-but-safe contrasts without altering forward propagation or supervised objectives. Across SGL, SimGCL, LightGCL, NCL, and DCCF on standard benchmarks, the module consistently improves Top-K ranking, demonstrating that principled negative construction complements diverse contrastive backbones rather than competing with their view designs. Sensitivity studies reveal broad plateaus for temperature, compact knee points for embedding size, and shallow optima for depth, yielding robust defaults and low tuning cost. Additional analyses further validate EMM's internal safeguards: ID exclusion and a cosine-similarity window mitigate false negatives introduced by hard-negative retrieval, and their combination consistently yields the best or tied-best performance. Limitations include leaving positive construction unchanged and focusing on offline evaluation. Future work will integrate popularity-aware debiasing and deployment-oriented efficiency improvements, such as more efficient ANN indexing and rebuild amortization to further strengthen practical deployment.

CRediT authorship contribution statement

Zhisheng Meng: Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Software, Supervision, Validation, Writing – original draft; **Jian Wang:** Formal analysis, Project administration, Resources, Supervision; **Lei Li:** Data curation, Formal analysis, Investigation, Validation; **Wenxia Chen:** Data curation, Investigation, Project administration, Resources; **Ziang He:** Conceptualization, Data curation, Investigation, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Bai, Z., Zheng, Y., Yang, P., Liu, S., Zhang, Y., & Chang, Y. (2025). Dcrlrec: Dual-domain contrastive reinforcement large language model for recommendation. *Information Processing & Management*, *62*(4), 104140.

Cai, M., Chen, L., Wang, Y., Bai, H., Sun, P., Wu, L., Zhang, M., & Wang, M. (2024). Popularity-aware alignment and contrast for mitigating popularity bias. In *Proceedings of the 30th ACM SIGKDD conference on knowledge discovery and data mining* KDD '24 (pp. 187–198). ACM. https://doi.org/10.1145/3637528.3671824

Cai, X., Huang, C., Xia, L., & Ren, X. (2023). LightGCL: Simple yet effective graph contrastive learning for recommendation. In *International conference on learning representations (ICLR)*.

Chuang, C.-Y., Robinson, J., Lin, Y.-C., Torralba, A., & Jegelka, S. (2020). Debiased contrastive learning. In *Advances in neural information processing systems (neurIPS)*.

Cremonesi, P., Koren, Y., & Turrin, R. (2010). Performance of recommender algorithms on top-N recommendation tasks. In *Proceedings of the fourth ACM conference on recommender systems (recsys '10)*.

Hao, Z., Xin, H., Wei, L., Tang, L., Wang, R., & Nie, F. (2024). Towards expansive and adaptive hard negative mining: Graph contrastive learning via subspace preserving. In *Proceedings of the ACM web conference 2024* WWW '24 (pp. 322–333). ACM. https://doi.org/10.1145/3589334.3645327

He, K., Fan, H., Wu, Y., Xie, S., & Girshick, R. (2020a). Momentum contrast for unsupervised visual representation learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)*.

He, L., Yan, W., Yi, T., & Sun, H. (2025). Self-supervised graph neural sequential recommendation with disentangling long and short-term interest. *ACM Transactions on Recommender Systems*, *4*(1), 1–25. https://doi.org/10.1145/3723173

He, X., Deng, K., Wang, X., Li, Y., Zhang, Y., & Wang, M. (2020b). LightGCN: Simplifying and powering graph convolution network for recommendation. In *Proceedings of the 43rd international ACM SIGIR conference on research and development in information retrieval* SIGIR '20 (pp. 639–648). ACM. https://doi.org/10.1145/3397271.3401063

Järvelin, K., & Kekäläinen, J. (2002). Cumulated gain-based evaluation of IR techniques. *Information Retrieval*.

Johnson, J., Douze, M., & Jégou, H. (2017). Billion-scale similarity search with GPUs. In *Proceedings of the 45th international ACM SIGIR conference on research and development in information retrieval*.

Kalantidis, Y., Sariyildiz, M. B., Pion, N., Weinzaepfel, P., & Larlus, D. (2020). Hard negative mixing for contrastive learning. In *Advances in neural information processing systems (neurIPS)*.

Keriven, N. (2022). Not too little, not too much: A theoretical analysis of graph (over)smoothing. In *Advances in neural information processing systems (neurIPS)*.

Kingma, D. P., & Ba, J. (2015). Adam: A method for stochastic optimization. In *International conference on learning representations (ICLR)*.

Lai, Y., Su, Y., Wei, L., He, T., Wang, H., Chen, G., Zha, D., Liu, Q., & Wang, X. (2024). Disentangled contrastive hypergraph learning for next POI recommendation. In *Proceedings of the 47th international ACM SIGIR conference on research and development in information retrieval* SIGIR 2024 (pp. 1452–1462). ACM. https://doi.org/10.1145/3626772.3657726

Lan, W., Zhou, G., Chen, Q., Wang, W., Pan, S., Pan, Y., & Zhang, S. (2024). Contrastive clustering learning for multi-behavior recommendation. *ACM Transactions on Information Systems*, *43*(1), 1–23. https://doi.org/10.1145/3698192

Li, H., Wang, Y., Xiao, Z., Yang, J., Zhou, C., Zhang, M., & Ju, W. (2025). Disco: Graph-based disentangled contrastive learning for cold-start cross-domain recommendation. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 12049–12057). (*vol. 39*).

Lin, Z., Tian, C., Hou, Y., & Zhao, W. X. (2022). Improving graph collaborative filtering with neighborhood-enriched contrastive learning. In *Proceedings of the ACM web conference 2022* WWW '22 (pp. 2320–2329). ACM. https://doi.org/10.1145/3485447.3512104

Liu, Y., Xia, L., & Huang, C. (2024). SelfGNN: Self-supervised graph neural networks for sequential recommendation. In *Proceedings of the 47th international ACM SIGIR conference on research and development in information retrieval* SIGIR 2024 (pp. 1609–1618). ACM. https://doi.org/10.1145/3626772.3657716

Liu, Z., Li, H., Chen, G., Ouyang, Y., Rong, W., & Xiong, Z. (2023). PopDCL: Popularity-aware debiased contrastive loss for collaborative filtering. In *Proceedings of the 32nd ACM international conference on information and knowledge management* CIKM '23 (pp. 1482–1492). ACM. https://doi.org/10.1145/3583780.3615009

Ni, J., Li, J., & McAuley, J. (2019). Justifying recommendations using distantly-labeled reviews and fine-grained aspects. In *Proceedings of the AAAI conference on artificial intelligence (AAAI)*.

Ni, X., Xiong, F., Zheng, Y., & Wang, L. (2024). Graph contrastive learning with kernel dependence maximization for social recommendation (CL-KDM). In *Proceedings of the ACM web conference 2024*.

Oono, K., & Suzuki, T. (2020). Graph neural networks exponentially lose expressive power for node classification. In *International conference on learning representations (ICLR)*.

Poole, B., Ozair, S., van den Oord, A., Alemi, A. A., & Tucker, G. (2019). On variational bounds of mutual information. In *Advances in neural information processing systems (neurIPS)*.

Ren, X., Xia, L., Yang, Y., Wei, W., Wang, T., Cai, X., & Huang, C. (2024). Sslrec: A self-supervised learning framework for recommendation. In *Proceedings of the 17th ACM international conference on web search and data mining* WSDM '24 (pp. 567–575). ACM. https://doi.org/10.1145/3616855.3635814

Ren, X., Xia, L., Zhao, J., Yin, D., & Huang, C. (2023). Disentangled contrastive collaborative filtering. In *Proceedings of the 46th international ACM SIGIR conference on research and development in information retrieval* SIGIR '23 (pp. 1137–1146). ACM. https://doi.org/10.1145/3539618.3591665

Sheng, X., Yuan, H., Zhao, P., Liu, G., Zhuang, F., & Sheng, V. (2024). Icsrec: Intent contrastive learning with cross subsequences for sequential recommendation. In *Proceedings of the 17th ACM international conference on web search and data mining (WSDM)*.

Tang, J., Dai, S., Sun, Z., Chen, X., Xu, J., Yu, W., Hu, L., Jiang, P., & Li, H. (2024). Towards robust recommendation via decision boundary-aware graph contrastive learning. In *Proceedings of the 30th ACM SIGKDD conference on knowledge discovery and data mining* KDD '24 (pp. 2854–2865). ACM. https://doi.org/10.1145/3637528.3671661

van den Oord, A., Li, Y., & Vinyals, O. (2018). Representation learning with contrastive predictive coding. arXiv:1807.03748.

Wang, C., Ma, W., Chen, C., Zhang, M., Liu, Y., & Ma, S. (2023). Sequential recommendation with multiple contrast signals. *ACM Transactions on Information Systems*, *41*(1), 1–27. https://doi.org/10.1145/3522673

Wang, J., Yu, L., Zhang, W., Gong, Y., Xu, Y., Wang, B., Zhang, P., & Zhang, D. (2017). Irgan: A minimax game for unifying generative and discriminative information retrieval models. In *Proceedings of the 40th international ACM SIGIR conference on research and development in information retrieval* (pp. 515–524).

Wang, T., & Isola, P. (2020). Understanding contrastive representation learning through alignment and uniformity on the hypersphere. In *Proceedings of the 37th international conference on machine learning (ICML)*.

Wu, J., Wang, X., Feng, F., He, X., Chen, L., Lian, J., & Xie, X. (2021). Self-supervised graph learning for recommendation. In *Proceedings of the 44th international ACM SIGIR conference on research and development in information retrieval* SIGIR '21 (pp. 726–735). ACM. https://doi.org/10.1145/3404835.3462862

Wu, Z., Xiong, Y., Yu, S. X., & Lin, D. (2018). Unsupervised feature learning via non-parametric instance discrimination. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)*.

Yang, H., Chen, H., Zhang, S., Sun, X., Li, Q., Zhao, X., & Xu, G. (2023). Generating counterfactual hard negative samples for graph contrastive learning. In *Proceedings of the ACM web conference 2023* WWW '23 (pp. 621–629). ACM. https://doi.org/10.1145/3543507.3583499

Yu, J., Xia, X., Chen, T., Cui, L., Hung, N. Q. V., & Yin, H. (2023). XSimGCL: Towards extremely simple graph contrastive learning for recommendation. *IEEE Transactions on Knowledge and Data Engineering*. https://doi.org/10.1109/TKDE.2023.3288135

Yu, J., Yin, H., Xia, X., Chen, T., Cui, L., & Hung, N. Q. V. (2022). Are graph augmentations necessary?: Simple graph contrastive learning for recommendation. In *Proceedings of the 45th international ACM SIGIR conference on research and development in information retrieval* SIGIR '22 (pp. 1294–1303). ACM. https://doi.org/10.1145/3477495.3531937

Zhang, S., Chen, L., Wang, C., Li, S., & Xiong, H. (2024). Temporal graph contrastive learning for sequential recommendation. In *Proceedings of the AAAI conference on artificial intelligence (AAAI)*.

Zhou, Y., Zhao, P., Fang, J., Qu, J., Liu, G., Zhuang, F., Sheng, V. S., & Xiaofang (2024). Meta-optimized joint generative and contrastive learning for sequential recommendation. In *Proceedings of the IEEE international conference on data engineering (ICDE)*.

Zhuang, H., Zhang, W. E., Dong, G. C., Yang, J., & Quan, Z. (2024). Trainable hard negative examples in contrastive learning for unsupervised abstractive summarization. In *Findings of the association for computational linguistics: EACL 2024*.